



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	155	Media
TechCrunch AI	9	Media
HackerNews	5	Community
The Verge AI	5	Media
MIT Tech Review	1	Media

VERIFIED INTELLIGENCE — 2 stories

VERI

Show HN: Id-agent - Token efficient UUID alternative for AI agents

HackerNews

+1 source

90%

HN 36

Hi HN, I'm Malachy, the founder of Tracecast. This project lets you generate interactive data apps on top of your data, using a Cursor-style AI chat. It stitches together Marimo, LangGraph agents, and data warehouse query tools. It has an Apache 2.0 license. The initial use case...

<https://github.com/vostride/id-agent>

VERI

Google's AI future demands trust — and your personal data

The Verge AI

+1 source

80%

Google has big promises for its AI-powered future - and a lot of it depends on your trust. At I/O 2026, Google described a bunch of new tools that it claims will make your life easier. Gemini Spark, Google's always-on AI agent, can help organize an upcoming event, while Daily...

<https://www.theverge.com/tech/934172/google-io-gemini-ai-trust-personal-data>

CONFIRMED SIGNALS — 171 stories

CONF

Show HN: Forge - Guardrails take an 8B model from 53% to 99% on agentic tasks

HackerNews

60%

HN 415

TL;DR: logbox is an open-source tool that pipes dev server logs to a local sqlite db with `| logbox collect`. Give Claude Code access by running `claude mcp add logbox -- logbox serve`. I used to copy & paste logs into Claude Code when manually testing my...

<https://github.com/antoinezambelli/forge>

CONF

Show HN: Clawputer - A personal AI assistant with a real computer and memory

HackerNews

50%

HN 4

No summary — see link for full article.

<https://clawputer.app>

CONF

Show HN: claude-autopilot, autonomous dev pipeline with multi-model review

HackerNews

50%

HN 3

No summary — see link for full article.

<https://github.com/axledbetter/claude-autopilot>

CONF

Show HN: audio.observer - AI news jingles you didn't ask for

HackerNews

50%

HN 3

No summary — see link for full article.

<https://audio.observer/>

CONF

Google updates its Gemini app to take on ChatGPT and Claude at IO 2026

TechCrunch AI

50%

The updates signal Google's push to turn its Gemini app into an all-purpose AI hub rather than a stand-alone chatbot.

<https://techcrunch.com/2026/05/19/google-updates-its-gemini-app-to-take-on-chatgpt-and-...>

CONF

You can now talk to your Gmail inbox, as seen at Google IO 2026

TechCrunch AI 50%

Google expands Gmail's AI Inbox with conversational voice search, letting users ask Gemini to find buried email details.

<https://techcrunch.com/2026/05/19/you-can-now-talk-to-your-gmail-inbox-as-seen-at-googl...>

CONF

From teen hacker to Iron Dome researcher, this founder raised \$28M to fight AI phishing

TechCrunch AI 50%

Ocean, an agentic email security platform, claims its AI can thoroughly analyze the context of every incoming email to detect fraud and impersonation attempts.

<https://techcrunch.com/2026/05/19/from-teen-hacker-to-iron-dome-researcher-this-founder...>

0 CONF

Elon Musk said Sam Altman 'stole' a non-profit — but the trial showed he had similar aims

TechCrunch AI 50%

The jury's speedy decision to reject Elon Musk's lawsuit against the other founders of OpenAI and Microsoft confirmed what we saw in the courtroom: Musk's case was a weak one, in part because he waited so long to file it.

<https://techcrunch.com/2026/05/19/elon-musk-said-sam-altman-stole-a-non-profit-but-the-...>

1 CONF

OpenAI is making it easier to check if an image was made by their models

TechCrunch AI 50%

OpenAI announced two new measures to help detect AI generated imagery: joining the open C2PA standard and adding Google's SynthID to its products.

<https://techcrunch.com/2026/05/19/openai-is-making-it-easier-to-check-if-an-image-was-m...>

2 CONF

Google introduces Gemini Spark, a 24/7 agentic assistant with Gmail integration, at IO 2026

TechCrunch AI 50%

At the Google I/O developer conference, the company announced a new agentic personal assistant called Gemini Spark, built from Gemini's base models and an agentic harness from Google Antigravity.

<https://techcrunch.com/2026/05/19/google-introduces-gemini-spark-a-24-7-agentic-assista...>

3 CONF

Google's Gemini Omni turns images, audio, and text into video — and that's just the start

TechCrunch AI 50%

Google's Gemini Omni is a new multimodal model that reasons across text, images, audio, and video to generate and edit videos through simple conversation — starting with Omni Flash.

<https://techcrunch.com/2026/05/19/googles-gemini-omni-turns-images-audio-and-text-into-...>

4 CONF

OpenAI co-founder Andrej Karpathy joins Anthropic's pre-training team

TechCrunch AI 50%

Pre-training is responsible for the large-scale training runs that give Claude its core knowledge and capabilities, according to the company. It's also one of the most expensive, compute-intensive phases of building a frontier model.

<https://techcrunch.com/2026/05/19/openai-co-founder-andrej-karpathy-joins-anthropics-pr...>

5 CONF

Here's why Elon Musk lost his suit against OpenAI

MIT Tech Review 50%

On Monday, the jury in Musk v. Altman dealt Elon Musk a major blow—reaching a unanimous advisory verdict that he had sued OpenAI too late and, as a result, his claims are barred by the applicable statutes of limitations. US District Judge Yvonne Gonzalez Rogers immediately...

<https://www.technologyreview.com/2026/05/18/1137488/elon-musk-suit-openai-verdict/>

6 CONF

Demis Hassabis said this might be the 'foothills of the singularity.' What?

The Verge AI 50%

Welcome to a "profound moment for humanity," according to Google DeepMind CEO Demis Hassabis, who closed out Google I/O's keynote presentation on Tuesday, saying: Google's cutting-edge research and products will help unlock AGI's incredible potential for the benefit of the...

<https://www.theverge.com/tech/934260/google-io-ai-singularity-demis-hassabis>

7 CONF

Gemini will use Volvo's external cameras to interpret parking signs

The Verge AI 50%

Gemini is gaining the power of sight and mobility. Today at the I/O conference, Google and Volvo announced that the AI-powered assistant will be able to access external cameras in the upcoming EX60 SUV to help explain and interpret its surroundings to vehicle owners. The upgrade...

<https://www.theverge.com/transportation/933556/google-io-gemini-volvo-ex60-camera-ai-pa...>

8 CONF

Google wants to compete with Anthropic's Mythos

The Verge AI 50%

Google is making a big push into cybersecurity. At I/O, the company announced that it was inviting select groups of experts to test the API for CodeMender, an "AI agent for code security" it debuted last October. The difference is that Google is now making the tool more widely...

<https://www.theverge.com/tech/933921/google-wants-to-compete-with-anthropics-mythos>

9 CONF

Gmail is going to start talking to you

The Verge AI 50%

Google is launching a big new feature for Gmail called Gmail Live, a new AI-powered voice mode that's basically the Gemini Live experience but built specifically for your inbox. To use Gmail Live, tap an icon that will appear in your search bar and just start talking. In a press...

<https://www.theverge.com/tech/932973/google-gmail-live-ai-keep-docs-io-2026>

0 CONF

Position: Let's Develop Data Probes to Fundamentally Understand How Data Affects LLM Performance

ArXiv CS.AI 50%

arXiv:2605.18801v1 Announce Type: new Abstract: Data is fundamental to large language models (LLMs). However, understanding of what makes certain data useful for different stages of an LLM workflow, including training, tuning, alignment, in-context learning, etc., and why,...

<https://arxiv.org/abs/2605.18801>

1 CONF

Operationalizing Document AI: A Microservice Architecture for OCR and LLM Pipelines in Production

ArXiv CS.AI 50%

arXiv:2605.18818v1 Announce Type: new Abstract: Academic research tends to focus on new models for document understanding creating a wide gap in the literature between model definition and running models at production scale. To close that gap, we present a microservice...

<https://arxiv.org/abs/2605.18818>

2 CONF

Evaluating the Utility of Personal Health Records in Personalized Health AI

ArXiv CS.AI 50%

arXiv:2605.18937v1 Announce Type: new Abstract: Patient-managed Personal Health Records (PHRs) promises to empower patients to better understand their health; but information in the record is complex, potentially hindering insights. In this study, we assess the potential of...

<https://arxiv.org/abs/2605.18937>

3 CONF

Learn-by-Wire Training Control Governance: Bounded Autonomous Training Under Stress for Stability and Efficiency

ArXiv CS.AI 50%

arXiv:2605.19008v1 Announce Type: new Abstract: Modern language-model training is increasingly exposed to instability, degraded runs, and wasted compute, especially under aggressive learning-rate, scale, and runtime-stress conditions. This paper introduces Learn-by-Wire Guard...

<https://arxiv.org/abs/2605.19008>

4 CONF

Embedding by Elicitation: Dynamic Representations for Bayesian Optimization of System Prompts

ArXiv CS.AI 50%

arXiv:2605.19093v1 Announce Type: new Abstract: System prompts are a central control mechanism in modern AI systems, shaping behavior across conversations, tasks, and user populations. Yet they are difficult to tune when feedback is available only as aggregate metrics rather...

<https://arxiv.org/abs/2605.19093>

5 CONF

C-ReD: A Comprehensive Chinese Benchmark for AI-Generated Text Detection Derived from Real-World Prompts

ArXiv CS.AI 50%

arXiv:2604.11796v2 Announce Type: replace-cross Abstract: Recently, large language models (LLMs) are capable of generating highly fluent textual content. While they offer significant convenience to humans, they also introduce various risks, like phishing and academic dishonesty....

<https://arxiv.org/abs/2604.11796>

6 CONF

Learning to Hand Off: Provably Convergent Workflow Learning under Interface Constraints

ArXiv CS.AI 50%

arXiv:2605.19140v1 Announce Type: new Abstract: We study workflow learning in a setting where specialized agents hand off control through a shared artifact, each agent observes only a local function of that artifact and its own private state, and no centralized learner accesses...

<https://arxiv.org/abs/2605.19140>

7 CONF

Hallucination as Exploit: Evidence-Carrying Multimodal Agents

ArXiv CS.AI 50%

arXiv:2605.19192v1 Announce Type: new Abstract: Multimodal agents use screenshots, documents, and webpages to choose tool calls. When a false visual claim triggers a click, email, extraction, or transfer, hallucination becomes an authorization failure rather than an...

<https://arxiv.org/abs/2605.19192>

8 CONF

Not all uncertainty is alike: volatility, stochasticity, and exploration

ArXiv CS.AI 50%

arXiv:2605.19215v1 Announce Type: new Abstract: Adaptive decision-making in biological and artificial intelligence requires balancing the exploitation of known outcomes with the exploration of uncertain alternatives. Although prior work suggests that uncertainty generally...

<https://arxiv.org/abs/2605.19215>

9 CONF

Can Large Language Models Revolutionize Survey Research? Experiments with Disaster Preparedness Responses

ArXiv CS.AI 50%

arXiv:2605.19229v1 Announce Type: new Abstract: Survey research faces mounting structural challenges: declining response rates, sample bias, block-wise missingness among at-risk respondents, and AI-assisted fraudulent completions in online panels. Large language models (LLMs)...

<https://arxiv.org/abs/2605.19229>

10 CONF

Agentic Trading: When LLM Agents Meet Financial Markets

ArXiv CS.AI 50%

arXiv:2605.19337v1 Announce Type: new Abstract: A growing body of work explores how Large Language Models (LLMs) can be embedded in trading systems as agents that perceive market information, retrieve context, reason about decisions, emit tradable actions, and adapt under market...

<https://arxiv.org/abs/2605.19337>

1 CONF

Generative Recursive Reasoning

ArXiv CS.AI 50%

arXiv:2605.19376v1 Announce Type: new Abstract: How should future neural reasoning systems implement extended computation? Recursive Reasoning Models (RRMs) offer a promising alternative to autoregressive sequence extension by performing iterative latent-state refinement with...

<https://arxiv.org/abs/2605.19376>

2 CONF

Generative Auto-Bidding with Unified Modeling and Exploration

ArXiv CS.AI 50%

arXiv:2605.19457v1 Announce Type: new Abstract: Automated bidding is central to modern digital advertising. Early rule-based methods lacked adaptability, while subsequent Reinforcement Learning approaches modeled bidding as a Markov Decision Process but struggled with long-term...

<https://arxiv.org/abs/2605.19457>

3 CONF

ReCrit: Transition-Aware Reinforcement Learning for Scientific Critic Reasoning

ArXiv CS.AI 50%

arXiv:2605.18799v1 Announce Type: cross Abstract: Large language models can fail in critic interaction not only by answering incorrectly, but also by abandoning an initially correct scientific solution after user criticism. This is especially risky in scientific reasoning, where...

<https://arxiv.org/abs/2605.18799>

4 CONF

Position: The Turing-Completeness of Real-World Autoregressive Transformers Relies Heavily on Context Management

ArXiv CS.AI 50%

arXiv:2605.19514v1 Announce Type: new Abstract: Many works make the eye-catching claim that Transformers are Turing-complete. However, the literature often conflates two distinct settings: (i) a fixed Transformer system setting, in which a fixed autoregressive Transformer is...

<https://arxiv.org/abs/2605.19514>

5 CONF

CounterFlow: A Two-Phase Inference-Time Sampling for Counterfactual Video Foley Generation

ArXiv CS.AI 50%

arXiv:2605.18916v1 Announce Type: cross Abstract: We investigate Counterfactual Video Foley Generation, which aims to adopt a sound-source identity that contradicts the visual evidence while remaining temporally synchronized to a silent video. Existing Video&Text-to-Audio (VT2A)...

<https://arxiv.org/abs/2605.18916>

6 CONF

Generative-Evaluative Agreement: A Necessary Validity Criterion for LLM-Enabled Adaptive Assessment

ArXiv CS.AI 50%

arXiv:2605.19529v1 Announce Type: new Abstract: When the same LLM generates assessment items, simulates student responses, and scores them, the validation loop is self-referential. We introduce Generative-Evaluative Agreement (GEA), a validity criterion measuring whether an...

<https://arxiv.org/abs/2605.19529>

7 CONF

Library Drift: Diagnosing and Fixing a Silent Failure Mode in Self-Evolving LLM Skill Libraries

ArXiv CS.AI 50%

arXiv:2605.19576v1 Announce Type: new Abstract: Self-evolving skill libraries face a silent failure mode we term \emph{library drift}: unbounded skill accumulation without outcome-driven lifecycle management causes retrieval degradation, false-positive injections, and...

<https://arxiv.org/abs/2605.19576>

8 CONF

Towards Multi-Model LLM Schedulers: Empirical Insights into Offloading and Preemption

ArXiv CS.AI 50%

arXiv:2605.19593v1 Announce Type: new Abstract: Modern deployments of Large Language Models (LLMs) increasingly require serving multiple models with diverse architectures, sizes, and specialization on shared, heterogeneous hardware. This setting introduces new challenges for...

<https://arxiv.org/abs/2605.19593>

9 CONF

When Tabular Foundation Models Meet Strategic Tabular Data: A Prior Alignment Approach

ArXiv CS.AI 50%

arXiv:2605.19662v1 Announce Type: new Abstract: Tabular foundation models based on pretrained prior-data fitted networks~(PFNs) have shown strong generalization on diverse tabular tasks, but they are typically designed for \emph{non-strategic} settings where data distributions...

<https://arxiv.org/abs/2605.19662>

0 CONF

Rotation-Aligned Key Channel Pruning for Efficient Vision-Language Model Inference

ArXiv CS.AI 50%

arXiv:2605.19218v1 Announce Type: cross Abstract: Vision-Language Models suffer severe KV cache pressure at inference, as a single image often encodes into thousands of tokens. Most existing methods exploit token sparsity through token pruning, but permanently discarding visual...

<https://arxiv.org/abs/2605.19218>

1 CONF

The World Won't Stay Still: Programmable Evolution for Agent Benchmarks

ArXiv CS.AI 50%

arXiv:2603.05910v2 Announce Type: replace Abstract: LLM-powered tool-calling agents fulfill user requests by interacting with environments, querying data, and invoking tools in a multi-turn process. Yet, most existing benchmarks evaluate these systems under static environment...

<https://arxiv.org/abs/2603.05910>

2 CONF

BuildArena: A Physics-Aligned Interactive Benchmark of LLMs for Engineering Construction

ArXiv CS.AI 50%

arXiv:2510.16559v5 Announce Type: replace Abstract: Engineering construction automation aims to transform natural language specifications into physically viable structures, requiring complex integrated reasoning under strict physical constraints. While modern LLMs possess broad...

<https://arxiv.org/abs/2510.16559>

3 CONF

CutVerse: A Compositional GUI Agents Benchmark for Media Post-Production Editing

ArXiv CS.AI 50%

arXiv:2605.19484v1 Announce Type: cross Abstract: While GUI agents have made significant progress in web navigation and basic operating system tasks, their capabilities in professional creative workflows remain largely underexplored. To bridge this gap, we introduce Cutverse, a...

<https://arxiv.org/abs/2605.19484>

4 CONF

GroupAffect-4: A Multimodal Dataset of Four-Person Collaborative Interaction

ArXiv CS.AI 50%

arXiv:2605.19765v1 Announce Type: new Abstract: Existing affective-computing, social-signal-processing, and meeting corpora capture important parts of human interaction, but they rarely support analysis of affect in co-located groups as a coupled individual, interpersonal, and...

<https://arxiv.org/abs/2605.19765>

5 CONF

BalanceRAG: Joint Risk Calibration for Cascaded Retrieval-Augmented Generation

ArXiv CS.AI 50%

arXiv:2605.20084v1 Announce Type: cross Abstract: Large language models (LLMs) can enhance factuality via retrieval-augmented generation (RAG), but applying RAG to every query is unnecessary when the model-only answer is reliable. This motivates cascaded RAG: each query is first...

<https://arxiv.org/abs/2605.20084>

6 CONF

Prior Knowledge or Search? A Study of LLM Agents in Hardware-Aware Code Optimization

ArXiv CS.AI 50%

arXiv:2605.19782v1 Announce Type: new Abstract: LLM discovery and optimization systems are increasingly applied across domains, implementing a common propose-evaluate-revise loop. Such optimization or discovery progresses via context conditioning on received feedback from an...

<https://arxiv.org/abs/2605.19782>

7 CONF

Explainable Wastewater Digital Twins: Adaptive Context-Conditioned Structured Simulators with Self-Falsifying Decision Support

ArXiv CS.AI 50%

arXiv:2605.19826v1 Announce Type: new Abstract: Operators of safety-critical industrial processes increasingly rely on digital twins to screen control interventions, but such simulators rarely carry certified safety guarantees. Wastewater treatment plants exemplify the gap:...

<https://arxiv.org/abs/2605.19826>

8 CONF

ContextFlow: Hierarchical Task-State Alignment for Long-Horizon Embodied Agents

ArXiv CS.AI 50%

arXiv:2605.19314v1 Announce Type: cross Abstract: Long-horizon embodied agents increasingly delegate navigation, search, approach, and manipulation to specialist executors. As these executors become stronger, the main bottleneck shifts from local skill execution to maintaining a...

<https://arxiv.org/abs/2605.19314>

9 CONF

Robotics-Inspired Guardrails for Foundation Models in Socially Sensitive Domains

ArXiv CS.AI 50%

arXiv:2605.19940v1 Announce Type: new Abstract: Foundation models are increasingly deployed in socially sensitive domains such as education, mental health, and caregiving, where failures are often cumulative and context-dependent. Existing guardrail approaches -- ranging from...

<https://arxiv.org/abs/2605.19940>

0 CONF

AutoResearchClaw: Self-Reinforcing Autonomous Research with Human-AI Collaboration

ArXiv CS.AI 50%

arXiv:2605.20025v1 Announce Type: new Abstract: Automating scientific discovery requires more than generating papers from ideas. Real research is iterative: hypotheses are challenged from multiple perspectives, experiments fail and inform the next attempt, and lessons accumulate...

<https://arxiv.org/abs/2605.20025>

1 CONF

Probing Embodied LLMs: When Higher Observation Fidelity Hurts Problem Solving

ArXiv CS.AI 50%

arXiv:2605.20072v1 Announce Type: new Abstract: Large Language Models are increasingly proposed as cognitive components for robotic systems, yet their opaque decision processes make it difficult to explain success or failure in closed-loop embodied tasks. Following an empirical...

<https://arxiv.org/abs/2605.20072>

2 **CONF****Neurosymbolic Learning for Inference-Time Argumentation****ArXiv CS.AI 50%**

arXiv:2605.20098v1 Announce Type: new Abstract: Claim verification is an important problem in high-stakes settings, including health and finance. When information underpinning claims is incomplete or conflicting, uncertain answers may be more appropriate than binary true or...

<https://arxiv.org/abs/2605.20098>

3 **CONF****Using Aristotle API for AI-Assisted Theorem Proving in Lean 4: A Formalisation Case Study of the Grasshopper Problem****ArXiv CS.AI 50%**

arXiv:2605.20120v1 Announce Type: new Abstract: AI-assisted theorem proving can now generate substantial Lean developments for olympiad-level mathematics, but the evidential status of such developments depends on which declarations are actually verified. This paper reports a...

<https://arxiv.org/abs/2605.20120>

4 **CONF****A Methodology for Selecting and Composing Runtime Architecture Patterns for Production LLM Agents****ArXiv CS.AI 50%**

arXiv:2605.20173v1 Announce Type: new Abstract: Production LLM agents combine stochastic model outputs with deterministic software systems, yet the boundary between the two is rarely treated as a first-class architectural object. This paper names that boundary the...

<https://arxiv.org/abs/2605.20173>

5 **CONF****Interceptive Divergence in Aesthetic Evaluation and Implications for Human-AI Alignment****ArXiv CS.AI 50%**

arXiv:2605.18759v1 Announce Type: cross Abstract: Artificial intelligence (AI), exemplified by large language models (LLMs), is rapidly approaching and in some cases surpassing human performance across a wide range of cognitive tasks. However, human nature is not limited to...

<https://arxiv.org/abs/2605.18759>

6 **CONF****Query-Conditioned Graph Retrieval for Contextualized LLM Reasoning in Personalized Wearable Data****ArXiv CS.AI 50%**

arXiv:2605.18763v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly applied to analyzing wearable sensing data, which are long-term, multimodal, and highly personalized. A key challenge is context selection: providing insufficient context limits...

<https://arxiv.org/abs/2605.18763>

7 **CONF****From Refusal to Recovery: A Control-Theoretic Approach to Generative AI Guardrails****ArXiv CS.AI 50%**

arXiv:2510.13727v2 Announce Type: replace Abstract: Generative AI systems are increasingly assisting and acting on behalf of end users in practical settings, from digital shopping assistants to next-generation autonomous cars. In this context, safety is no longer about blocking...

<https://arxiv.org/abs/2510.13727>

- 8 **CONF**
ClusterRAG: Cluster-Based Collaborative Filtering for Personalized Retrieval-Augmented Generation
 ArXiv CS.AI 50%
 arXiv:2605.18769v1 Announce Type: cross Abstract: Personalized Retrieval-Augmented Generation (RAG) relies on accurately selecting user-relevant documents. In practice, existing RAG approaches often suffer from high retrieval costs and overlook that collaborative signals from...
<https://arxiv.org/abs/2605.18769>
- 9 **CONF**
Decentralized autonomous organization and blockchain-based incentivization framework for community-based facilities management
 ArXiv CS.AI 50%
 arXiv:2605.18773v1 Announce Type: cross Abstract: Traditional facility management often relies on centralized decision-making structures that limit stakeholder participation, leading to misalignment with occupant needs and reduced satisfaction. This paper proposes a novel...
<https://arxiv.org/abs/2605.18773>
- 0 **CONF**
Artificial Phantasia: Emergent Mental Imagery in Large Language Models
 ArXiv CS.AI 50%
 arXiv:2509.23108v2 Announce Type: replace Abstract: Can visual imagery be driven solely by language? This idea goes against cognitive science's traditional view that visual mental imagery is only possible through pictorial representations. Large Language Models (LLMs) provide...
<https://arxiv.org/abs/2509.23108>
- 1 **CONF**
Query-Aware Flow Diffusion for Graph-Based RAG with Retrieval Guarantees
 ArXiv CS.AI 50%
 arXiv:2605.18775v1 Announce Type: cross Abstract: Graph-based Retrieval-Augmented Generation (RAG) systems leverage interconnected knowledge structures to capture complex relationships that flat retrieval struggles with, enabling multi-hop reasoning. Yet most existing...
<https://arxiv.org/abs/2605.18775>
- 2 **CONF**
Mask-to-Correct⁺: Leveraging Retriever Diversity for Masking-guided Faithful Fact Correction
 ArXiv CS.AI 50%
 arXiv:2605.18776v1 Announce Type: cross Abstract: The rapid spread of misinformation on social media highlights the need for robust, automated fact correction frameworks. However, existing works rely on supervised learning from manually annotated claim-evidence pairs, which are...
<https://arxiv.org/abs/2605.18776>
- 3 **CONF**
A Reproducibility Analysis of PO4ISR: Diagnosing and Mitigating Semantic Drift in LLM-Based Session Recommendation
 ArXiv CS.AI 50%
 arXiv:2605.18780v1 Announce Type: cross Abstract: Reasoning-based Large Language Models (LLMs) like PO4ISR have set new benchmarks in session-based recommendation. However, the reproducibility of their reasoning capabilities across diverse semantic domains remains unexplored. In...
<https://arxiv.org/abs/2605.18780>

4 **CONF****Can LLMs Emulate Human Belief Dynamics?****ArXiv CS.AI** **50%**

arXiv:2605.18781v1 Announce Type: cross Abstract: Can LLMs simulate how humans form and change beliefs in social networks? We put this to the test by replicating an established study on belief dynamics, evaluating 12 LLMs across multiple model families and parameter sizes. The...

<https://arxiv.org/abs/2605.18781>

5 **CONF****The Insurability Frontier of AI Risk: Mapping Threats to Affirmative Coverage, Silent Exposures, and Exclusions****ArXiv CS.AI** **50%**

arXiv:2605.18784v1 Announce Type: cross Abstract: The rapid diffusion of agentic AI has created a new coverage problem for commercial insurance: some AI-mediated losses are now affirmatively insured, some create silent-AI exposure under legacy cyber, technology...

<https://arxiv.org/abs/2605.18784>

6 **CONF****Robust Basis Spline Decoupling for the Compression of Transformer Models****ArXiv CS.AI** **50%**

arXiv:2605.18794v1 Announce Type: cross Abstract: Decoupling is a powerful modeling paradigm for representing multivariate functions as compositions of linear transformations and univariate nonlinear functions. A single-layer decoupling can be viewed as a fully connected neural...

<https://arxiv.org/abs/2605.18794>

7 **CONF****RecoAtlas: From Semantic Plausibility to Set-Level Utility in LLM Recommendation Agents****ArXiv CS.AI** **50%**

arXiv:2605.18805v1 Announce Type: cross Abstract: LLM recommendation agents increasingly produce structured recommendation reports: sets of items accompanied by natural-language justifications. Yet existing evaluations often reduce this setting to reranking small shortlisted...

<https://arxiv.org/abs/2605.18805>

8 **CONF****Compositional Literary Primitives in Instruction-Tuned LLMs: Cross-Architectural SAE Features for Self, Style, and Affect****ArXiv CS.AI** **50%**

arXiv:2605.18808v1 Announce Type: cross Abstract: We characterize a compositional architecture of literary primitives in two instruction-tuned large language models (Llama 3.1 8B-Instruct and Gemma 2 9B-IT) via sparse autoencoders on mid-depth residual streams. Four feature...

<https://arxiv.org/abs/2605.18808>

9 **CONF****Composition of Memory Experts for Diffusion World Models****ArXiv CS.AI** **50%**

arXiv:2605.18813v1 Announce Type: cross Abstract: World models aim to predict plausible futures consistent with past observations, a capability central to planning and decision-making in reinforcement learning. Yet, existing architectures face a fundamental memory trade-off:...

<https://arxiv.org/abs/2605.18813>

0 CONF

Hybrid-LoRA: Bridging Full Fine-Tuning and Low-Rank Adaptation for Post-Training

ArXiv CS.AI 50%

arXiv:2605.18822v1 Announce Type: cross Abstract: Post-training has become essential for adapting large language models (LLMs) to complex downstream behaviors, including instruction following, preference alignment, and multi-step reasoning. Reinforcement learning with verifiable...

<https://arxiv.org/abs/2605.18822>

1 CONF

Quantifying the Pre-training Dividend: Generative versus Latent Self-Supervised Learning for Time Series Foundation Models

ArXiv CS.AI 50%

arXiv:2605.19462v1 Announce Type: cross Abstract: The success of self-supervised learning (SSL) in vision and NLP has motivated its rapid adoption for time series. However, research has focused primarily on Generative paradigms and forecasting tasks, leaving the broader utility...

<https://arxiv.org/abs/2605.19462>

2 CONF

Precision Tracked Transformer via Kalman Filtering, Kriging and Process Noise

ArXiv CS.AI 50%

arXiv:2605.18832v1 Announce Type: cross Abstract: The Transformer is the foundational building block of modern AI, yet offers no principled handling of \emph{uncertainty}, which is prevalent in real applications: cold-start tokens with sparse histories in sequential...

<https://arxiv.org/abs/2605.18832>

3 CONF

Automated Big Data Quality Assessment using Knowledge Graph Embeddings

ArXiv CS.AI 50%

arXiv:2605.18833v1 Announce Type: cross Abstract: Automated data quality assessment is crucial for managing big data, but existing solutions face challenges in achieving accurate context-aware assessment. This paper presents a novel knowledge-based approach to enhance automated...

<https://arxiv.org/abs/2605.18833>

4 CONF

Lying Is Just a Phase: The Hidden Alignment Transition in Language Model Scaling

ArXiv CS.AI 50%

arXiv:2605.18838v1 Announce Type: cross Abstract: Scaling laws predict loss from compute but not how capabilities interact. We measure the coupling between reasoning and truthfulness across 63 base models from 16 families and find a regime change invisible to loss curves: below...

<https://arxiv.org/abs/2605.18838>

5 CONF

The Growing Pains of Frontier Models: When Leaderboards Stop Separating and What to Measure Next

ArXiv CS.AI 50%

arXiv:2605.18840v1 Announce Type: cross Abstract: Leaderboards rank frontier models on independent axes but do not reveal whether capabilities reinforce or trade off across releases -- and at the frontier, this interaction is the more informative signal. We decompose paired...

<https://arxiv.org/abs/2605.18840>

6 CONF

Graph-Driven Cross-Industry Real-Time Monitoring Framework for Anti-Money Laundering Detection in Converged Mobility-Energy Supply Chain Networks

ArXiv CS.AI 50%

arXiv:2605.18844v1 Announce Type: cross Abstract: With the deep integration of the travel and energy industries, cross-industry supply chain finance has gradually become a high-risk field of hidden money laundering incidents. For this reason, this work proposes a graph-driven...

<https://arxiv.org/abs/2605.18844>

7 CONF

Transformers Linearly Represent Highly Structured World Models

ArXiv CS.AI 50%

arXiv:2605.18847v1 Announce Type: cross Abstract: Do transformers, when trained on sequential reasoning traces, build internal models of the underlying task? And if so, does the structure of those internal representations mirror the structure of the domain? We train an 8-layer...

<https://arxiv.org/abs/2605.18847>

8 CONF

IR-Agent: Expert-Inspired LLM Agents for Structure Elucidation from Infrared Spectra

ArXiv CS.AI 50%

arXiv:2508.16112v2 Announce Type: replace Abstract: Spectral analysis provides crucial clues for the elucidation of unknown materials. Among various techniques, infrared spectroscopy (IR) plays an important role in laboratory settings due to its high accessibility and low cost...

<https://arxiv.org/abs/2508.16112>

9 CONF

TwinRouterBench: Fast Static and Live Dynamic Evaluation for Realistic Agentic LLM Routing

ArXiv CS.AI 50%

arXiv:2605.18859v1 Announce Type: cross Abstract: LLM routing matters most in long-horizon applications such as coding agents, deep research systems, and computer-use agents, where a single user request triggers many model calls. Routing each call to the cheapest sufficient...

<https://arxiv.org/abs/2605.18859>

0 CONF

Towards Family-Grouped Hierarchical Federated Learning on Sub-5KB Models: A Feasibility Study of Privacy-Preserving ECG Monitoring for Ultra-Resource-Constrained Wearables

ArXiv CS.AI 50%

arXiv:2605.18862v1 Announce Type: cross Abstract: Cardiovascular disease remains the leading cause of death worldwide, and early detection of arrhythmias through continuous ECG monitoring on wearable devices can prevent life-threatening events. Federated Learning (FL) enables...

<https://arxiv.org/abs/2605.18862>

1 CONF

TSR: Trajectory-Search Rollouts for Multi-Turn RL of LLM Agents

ArXiv CS.AI 50%

arXiv:2602.11767v3 Announce Type: replace Abstract: Advances in large language models (LLMs) are driving a shift toward using reinforcement learning (RL) to train agents from iterative, multi-turn interactions across tasks. However, multi-turn RL remains challenging as rewards...

<https://arxiv.org/abs/2602.11767>

2 **CONF****DarkLLM: Learning Language-Driven Adversarial Attacks with Large Language Models**

ArXiv CS.AI 50%

arXiv:2605.18868v1 Announce Type: cross Abstract: While vision and multimodal foundation models underpin critical tasks from perception to complex reasoning, they remain highly vulnerable to adversarial attacks. However, traditional adversarial attacks are typically limited to...

<https://arxiv.org/abs/2605.18868>

3 **CONF****Distributional Energy-Based Models for Uncertainty-Aware Structured LLM Reasoning**

ArXiv CS.AI 50%

arXiv:2605.18871v1 Announce Type: cross Abstract: When Large Language Models produce structured outputs such as travel plans, code solutions, or multi-step proofs, individual reasoning steps may appear correct while the output as a whole violates budgets, fails test cases, or...

<https://arxiv.org/abs/2605.18871>

4 **CONF****Multi-Scale Generative Modeling with Heat Dissipation Flow Matching**

ArXiv CS.AI 50%

arXiv:2605.19371v1 Announce Type: cross Abstract: Diffusion models are widely used in image generation, with most relying on noise-based corruption and denoising. A distinct branch instead uses blur as the main corruption, preserving better color budgets and multi-scale detail...

<https://arxiv.org/abs/2605.19371>

5 **CONF****To Call or Not to Call: Diagnosing Intrinsic Over-Calling Bias in LLM Agents**

ArXiv CS.AI 50%

arXiv:2605.18882v1 Announce Type: cross Abstract: LLM agents exhibit a consistent tendency to over-call, invoking tools even in situations where none is needed. On the When2Call benchmark, six models from three families show high call accuracy but much lower no-call accuracy,...

<https://arxiv.org/abs/2605.18882>

6 **CONF****Stop Drawing Scientific Claims from LLM Social Simulations Without Robustness Audits**

ArXiv CS.AI 50%

arXiv:2605.18890v1 Announce Type: cross Abstract: The scientific claims drawn from LLM social simulations should be no stronger than the robustness audits that support them. Generative agents bring new expressive power to agent-based modeling, enabling simulations of collective...

<https://arxiv.org/abs/2605.18890>

7 **CONF****Auditing Reasoning-Trace Memorization Claims after Unlearning with Head-Conditioned Canaries**

ArXiv CS.AI 50%

arXiv:2605.18891v1 Announce Type: cross Abstract: Evaluations of unlearning on reasoning models sometimes show a bypass pattern. The answer side looks unlearned, but the model's own thinking trace keeps emitting the forgotten content, and the gap is taken as evidence that the...

<https://arxiv.org/abs/2605.18891>

8 **CONF****Cross-Subject Intracranial EEG Reconstruction from Scalp Recordings Using Multi-Scale Cross-Attention Transformers****ArXiv CS.AI** **50%**

arXiv:2605.18897v1 Announce Type: cross Abstract: Intracranial EEG (iEEG) provides high-fidelity neural recordings essential for clinical and brain-computer interface applications, but acquiring these signals requires invasive surgery. While recent studies have attempted to...

<https://arxiv.org/abs/2605.18897>

9 **CONF****Don't Let Bandit Feedback Pull Continual LLM-Recommender Updates Off Target****ArXiv CS.AI** **50%**

arXiv:2605.18899v1 Announce Type: cross Abstract: Generative LLM-based recommenders (LLM-Rec) require continual post-deployment updates, yet deployment logs provide only policy-shaped contextual bandit feedback: outcomes are observed solely for items exposed by a prior serving...

<https://arxiv.org/abs/2605.18899>

0 **CONF****DMN: A Compositional Framework for Jailbreaking Multimodal LLMs with Multi-Image Inputs****ArXiv CS.AI** **50%**

arXiv:2605.18915v1 Announce Type: cross Abstract: Multimodal Large Language Models (MLLMs) are vulnerable to jailbreak attacks, which can elicit harmful responses from MLLMs. Many MLLMs support multi-image inputs, inadvertently introducing new vulnerabilities due to less efforts...

<https://arxiv.org/abs/2605.18915>

1 **CONF****SynGR: Unleashing the Potential of Cross-Modal Synergy for Generative Recommendation****ArXiv CS.AI** **50%**

arXiv:2605.18920v1 Announce Type: cross Abstract: Generative Recommendation (GR) has emerged as a promising paradigm by formulating item recommendation as a sequence-to-sequence generation task over item identifiers. Recent studies have incorporated multimodal signals to provide...

<https://arxiv.org/abs/2605.18920>

2 **CONF****OEP: Poisoning Self-Evolving LLM Agents via Locally Correct but Non-Transferable Experiences****ArXiv CS.AI** **50%**

arXiv:2605.18930v1 Announce Type: cross Abstract: Memory-augmented large language model (LLM) agents use iterative reflection and self-evolution to solve complex tasks, but these mechanisms introduce security risks. Existing agentic memory attacks require privileged access or...

<https://arxiv.org/abs/2605.18930>

3 **CONF****Markov Chain Decoders Overcome the Heavy-Tail Limitations of Lipschitz Generative Models****ArXiv CS.AI** **50%**

arXiv:2605.18931v1 Announce Type: cross Abstract: Heavy-tailed distributions are prevalent in performance evaluation, network traffic, and risk modeling. This behavior poses a fundamental challenge for modern deep generative models. Standard Variational Autoencoders (VAEs)...

<https://arxiv.org/abs/2605.18931>

4 **CONF****HypergraphFormer: Learning Hypergraphs from LLMs for Editable Floor Plan Generation**

ArXiv CS.AI 50%

arXiv:2605.18932v1 Announce Type: cross Abstract: In this work, we propose HypergraphFormer, a novel and efficient approach to floor plan generation based on learning hypergraph representations with a large language model (LLM). The model is trained via supervised fine-tuning to...

<https://arxiv.org/abs/2605.18932>

5 **CONF****Shaping the Prior: How Synthetic Task Distributions Determine Tabular Foundation Model Quality**

ArXiv CS.AI 50%

arXiv:2605.18971v1 Announce Type: cross Abstract: What determines the quality of a tabular foundation model? Unlike language or vision, tabular foundation models acquire their inductive biases almost entirely from synthetic pretraining distributions, yet the design of these...

<https://arxiv.org/abs/2605.18971>

6 **CONF****Surviving the Unseen: Predictive Defense for Novel Multi-Turn Multimodal Attacks**

ArXiv CS.AI 50%

arXiv:2605.18988v1 Announce Type: cross Abstract: The expansion of Multimodal Large Language Models (MLLMs) and their integration into autonomous agentic workflows has introduced a non-stationary attack surface. Empirical observations indicate that adversaries employ...

<https://arxiv.org/abs/2605.18988>

7 **CONF****RLFTSim: Realistic and Controllable Multi-Agent Traffic Simulation via Reinforcement Learning Fine-Tuning**

ArXiv CS.AI 50%

arXiv:2605.19033v1 Announce Type: cross Abstract: Supervised open-loop training has been widely adopted for training traffic simulation models; however, it fails to capture the inherently dynamic, multi-agent interactions common in complex driving scenarios. We introduce...

<https://arxiv.org/abs/2605.19033>

8 **CONF****Toward an AI-Powered Computational Testbed for Workforce Policy**

ArXiv CS.AI 50%

arXiv:2605.19064v1 Announce Type: cross Abstract: Workforce transformations are difficult to forecast and costly to mismanage. In particular, the integration of artificial intelligence into knowledge work currently affects a substantial share of the global workforce, yet this...

<https://arxiv.org/abs/2605.19064>

9 **CONF****Benchmarking Commercial ASR Systems on Code-Switching Speech: Arabic, Persian, and German**

ArXiv CS.AI 50%

arXiv:2605.19069v1 Announce Type: cross Abstract: Code-switching -- the natural alternation between two languages within a single utterance -- represents one of the most challenging and under-studied conditions for automatic speech recognition (ASR). Existing commercial ASR...

<https://arxiv.org/abs/2605.19069>

00 CONF

Learning Long-Term Temporal Dependencies in Photovoltaic Power Output Prediction Through Multi-Horizon Forecasting

ArXiv CS.AI 50%

arXiv:2605.19074v1 Announce Type: cross Abstract: The rapid global expansion of solar photovoltaic (PV) capacity-reaching a record 597 GW in 2024-highlights the urgent need for robust forecasting models to mitigate the grid instability caused by the intermittent nature of solar...

<https://arxiv.org/abs/2605.19074>

01 CONF

CRAFT: Critic-Refined Adaptive Key-Frame Targeting for Multimodal Video Question Answering

ArXiv CS.AI 50%

arXiv:2605.19075v1 Announce Type: cross Abstract: Grounded multi-video question answering over real-world news events requires systems to surface query-relevant evidence across heterogeneous video archives while attributing every claim to its supporting source. We introduce...

<https://arxiv.org/abs/2605.19075>

02 CONF

GOAL: Graph-based Objective-Aligned Diffusion Solvers for Dynamic Multi-Objective Optimization

ArXiv CS.AI 50%

arXiv:2605.19119v1 Announce Type: cross Abstract: Existing neural combinatorial optimization solvers frame solution search as imitation of optimal decisions, inherently limiting their utility to single-objective minimization and static constraints. We propose GOAL, a conditioned...

<https://arxiv.org/abs/2605.19119>

03 CONF

EgoBabyVLM: Benchmarking Cross-Modal Learning from Naturalistic Egocentric Video Data

ArXiv CS.AI 50%

arXiv:2605.19130v1 Announce Type: cross Abstract: Children acquire language grounding with remarkable robustness from limited visuo-linguistic input in ways that surpass today's best large multimodal models. Recent research suggests current vision-language models (VLMs) trained...

<https://arxiv.org/abs/2605.19130>

04 CONF

Be Kind, Rewrite: Benign Projections via Rewriting Defend Against LLM Data Poisoning Attacks

ArXiv CS.AI 50%

arXiv:2605.19147v1 Announce Type: cross Abstract: Large language models (LLMs) are highly susceptible to backdoor attacks (BAs), wherein training samples are poisoned using trigger-based harmful content. Furthermore, existing defenses have proven ineffective when extensively...

<https://arxiv.org/abs/2605.19147>

05 CONF

Quantized Machine Learning Models for Medical Imaging in Low-Resource Healthcare Settings

ArXiv CS.AI 50%

arXiv:2605.19207v1 Announce Type: cross Abstract: Deep learning models have shown strong performance in medical image analysis, but deploying them in low-resource clinical environments remains difficult due to computational, memory, and power constraints. This paper presents a...

<https://arxiv.org/abs/2605.19207>

06 CONF

Diagnosing Multi-step Reasoning Failures in Black-box LLMs via Stepwise Confidence Attribution

ArXiv CS.AI 50%

arXiv:2605.19228v1 Announce Type: cross Abstract: Large Language Models have achieved strong performance on reasoning tasks with objective answers by generating step-by-step solutions, but diagnosing where a multi-step reasoning trace might fail remains difficult. Confidence...

<https://arxiv.org/abs/2605.19228>

07 CONF

AI Technologies in Language Access: Attitudes Towards AI and the Human Value of Language Access Managers

ArXiv CS.AI 50%

arXiv:2605.19234v1 Announce Type: cross Abstract: The rapid emergence of AI technologies is reshaping translation practices and theory across the board. This paper deals with the impact of AI in language access. This area is characterized by the need to serve broad and diverse...

<https://arxiv.org/abs/2605.19234>

08 CONF

PhyWorld: Physics-Faithful World Model for Video Generation

ArXiv CS.AI 50%

arXiv:2605.19242v1 Announce Type: cross Abstract: World simulators can provide safe and scalable environments for training Physical AI systems before real-world deployment. Large video generation models are emerging as a promising basis for such simulators because they can...

<https://arxiv.org/abs/2605.19242>

09 CONF

Euclidean Embedding of Data Using Local Distances

ArXiv CS.AI 50%

arXiv:2605.19243v1 Announce Type: cross Abstract: We study the problem of recovering a globally consistent Euclidean embedding of data, given only a local distance graph and propose a method that optimally represents these distances. The method operates solely on a neighborhood...

<https://arxiv.org/abs/2605.19243>

10 CONF

Are Rationales Necessary and Sufficient? Tuning LLMs for Explainable Misinformation Detection

ArXiv CS.AI 50%

arXiv:2605.19285v1 Announce Type: cross Abstract: The rapid spread of misinformation on social media platforms has become a formidable challenge. To mitigate its proliferation, Misinformation Detection (MD) has emerged as a critical research topic. Traditional MD approaches...

<https://arxiv.org/abs/2605.19285>

11 CONF

Latent Laplace Diffusion for Irregular Multivariate Time Series

ArXiv CS.AI 50%

arXiv:2605.19805v1 Announce Type: cross Abstract: Irregular multivariate time series impose a trade-off for long-horizon forecasting: discrete methods can distort temporal structure via re-gridding, while continuous-time models often require sequential solvers prone to drift. To...

<https://arxiv.org/abs/2605.19805>

12 CONF

Brain alignment of reasoning and action representations from vision-language and action models during naturalistic gameplay

ArXiv CS.AI 50%

arXiv:2605.19352v1 Announce Type: cross Abstract: Understanding how humans and artificial intelligence systems predict and plan by interacting with their environment is a fundamental challenge at the intersection of neuroscience and machine learning. Most brain-encoding studies...

<https://arxiv.org/abs/2605.19352>

13 CONF

Conflict-Free Replicated Data Types for Neural Network Model Merging: A Two-Layer Architecture Enabling CRDT-Compliant Model Merging Across 26 Strategies

ArXiv CS.AI 50%

arXiv:2605.19373v1 Announce Type: cross Abstract: All 26 neural network merge strategies we tested including weight averaging, SLERP, TIES, DARE, Fisher merging, and evolutionary approaches -- fail the algebraic properties (commutativity, associativity, idempotency) required for...

<https://arxiv.org/abs/2605.19373>

14 CONF

Quantifying the Generalization Gap in Seizure Detection: A Large-Scale Empirical Benchmark via the SzCORE Challenge

ArXiv CS.AI 50%

arXiv:2505.18191v2 Announce Type: replace-cross Abstract: Reliable automatic seizure detection from long-term electroencephalography (EEG) remains an unsolved challenge, as current models often fail to generalize across patients or clinical settings. Manual EEG review still is...

<https://arxiv.org/abs/2505.18191>

15 CONF

Backtracking When It Strays: Mitigating Dual Exposure Biases in LLM Reasoning Distillation

ArXiv CS.AI 50%

arXiv:2605.19433v1 Announce Type: cross Abstract: Large language models (LLMs) have achieved remarkable success in complex reasoning tasks via long chain-of-thought (CoT), yet their immense computational overhead hinders real-world deployment. LLM reasoning distillation...

<https://arxiv.org/abs/2605.19433>

16 CONF

KappaPlace: Learning Hyperspherical Uncertainty for Visual Place Recognition via Prototype-Anchored Supervision

ArXiv CS.AI 50%

arXiv:2605.19435v1 Announce Type: cross Abstract: Visual Place Recognition (VPR) is critical for autonomous navigation, yet state-of-the-art methods lack well-calibrated uncertainty estimation. Standard pipelines cannot reliably signal when a query is ambiguous or a match is...

<https://arxiv.org/abs/2605.19435>

17 CONF

Rethinking the Design Space of Reinforcement Learning for Diffusion Models: On the Importance of Likelihood Estimation Beyond Loss Design

ArXiv CS.AI 50%

arXiv:2602.04663v2 Announce Type: replace-cross Abstract: Reinforcement learning has been widely applied to diffusion and flow models for visual tasks such as text-to-image generation. However, these tasks remain challenging because diffusion models have intractable likelihoods,...

<https://arxiv.org/abs/2602.04663>

18 CONF

CANINE: Coaching Visually Impaired Users for Interactive Navigation with a Robot Guide Dog

ArXiv CS.AI 50%

arXiv:2605.19501v1 Announce Type: cross Abstract: Robot guide dogs offer navigation assistance that greatly expands the independent mobility of the visually impaired, but their effective use requires subtle human-robot coordination that is difficult for users to learn from...

<https://arxiv.org/abs/2605.19501>

19 CONF

A novel YOLO26-MoE optimized by an LLM agent for insulator fault detection considering UAV images

ArXiv CS.AI 50%

arXiv:2605.19595v1 Announce Type: cross Abstract: The inspection of electrical power line insulators is essential for ensuring grid reliability and preventing failures caused by damaged or degraded insulation components. In recent years, Unmanned Aerial Vehicles (UAVs) combined...

<https://arxiv.org/abs/2605.19595>

20 CONF

The Accessibility Capability Boundary: Operational Limits and Expansion Potential of AI-Generated Browser-Native Accessibility Systems

ArXiv CS.AI 50%

arXiv:2605.19638v1 Announce Type: cross Abstract: As large language models (LLMs) demonstrate increasing competence in synthesizing functional user interfaces, a fundamental question emerges in accessibility computing: \textit{how far can AI-driven accessibility systems go?}...

<https://arxiv.org/abs/2605.19638>

21 CONF

CriterAlign: Criterion-Centric Rationale Alignment for Code Preference Judging

ArXiv CS.AI 50%

arXiv:2605.19665v1 Announce Type: cross Abstract: Pairwise human preference prediction is central to evaluating code-generation systems, where quality often depends on task-specific trade-offs beyond functional correctness. While rubric-based LLM judges improve interpretability...

<https://arxiv.org/abs/2605.19665>

22 CONF

Measuring Safety Alignment Effects in Autonomous Security Agents

ArXiv CS.AI 50%

arXiv:2605.19722v1 Announce Type: cross Abstract: Do stock safety-aligned language models and their uncensored or ablated derivatives behave differently when run as autonomous security agents? Single-turn refusal benchmarks cannot answer this question: security agents must...

<https://arxiv.org/abs/2605.19722>

23 CONF

LIFT and PLACE: A Simple, Stable, and Effective Knowledge Distillation Framework for Lightweight Diffusion Models

ArXiv CS.AI 50%

arXiv:2605.19729v1 Announce Type: cross Abstract: We demonstrate that in knowledge distillation for diffusion models, the teacher network's highly complex denoising process - stemming from its substantially larger capacity - poses a significant challenge for the student model to...

<https://arxiv.org/abs/2605.19729>

24 CONF

Operationalising Artificial Intelligence Bills of Materials (AIBOMs) for Verifiable AI Provenance and Lifecycle Assurance

ArXiv CS.AI 50%

arXiv:2605.19755v1 Announce Type: cross Abstract: Artificial Intelligence (AI) systems are increasingly dependent on complex, multi-layered software supply chains that introduce challenges for reproducibility, transparency, and security assurance. This study presents an...

<https://arxiv.org/abs/2605.19755>

25 CONF

AR1-ZO: Topology-Aware Rank-1 Zeroth-Order Queries for High-Rank LoRA Fine-Tuning

ArXiv CS.AI 50%

arXiv:2605.19767v1 Announce Type: cross Abstract: Zeroth-order (ZO) optimization enables large-language-model fine-tuning without storing backpropagation activations, while LoRA supplies compact trainable adapters. Combining them creates a rank paradox: increasing LoRA rank...

<https://arxiv.org/abs/2605.19767>

26 CONF

Synergistic Foundation Models for Semi-Supervised Fetal Cardiac Ultrasound Analysis: SAM-Med2D Boundary Refinement and DINOv3 Semantic Enhancement

ArXiv CS.AI 50%

arXiv:2605.19799v1 Announce Type: cross Abstract: We present a semi-supervised framework for joint segmentation and classification of fetal cardiac ultrasound images. Built upon the EchoCare multi-task backbone, our method integrates SAM-Med2D for boundary refinement and...

<https://arxiv.org/abs/2605.19799>

27 CONF

FLUXtrapolation: A benchmark on extrapolating ecosystem fluxes

ArXiv CS.AI 50%

arXiv:2605.19812v1 Announce Type: cross Abstract: We introduce FLUXtrapolation, a benchmark for extrapolating ecosystem fluxes under progressively harder distribution shifts. Ecosystem fluxes are central to understanding the carbon, water, and energy cycles, yet they can only be...

<https://arxiv.org/abs/2605.19812>

28 CONF

ST-TGExplainer: Disentangling Stability and Transition Patterns for Temporal GNN Interpretability

ArXiv CS.AI 50%

arXiv:2605.19822v1 Announce Type: cross Abstract: Temporal graph neural networks (TGNNs) have gained significant traction for solving real-world temporal graph tasks. However, their interpretability remains limited, as most TGNNs fail to identify which historical interactions...

<https://arxiv.org/abs/2605.19822>

29 CONF

FineBench: Benchmarking and Enhancing Vision-Language Models for Fine-grained Human Activity Understanding

ArXiv CS.AI 50%

arXiv:2605.19846v1 Announce Type: cross Abstract: Vision-Language Models (VLMs) have demonstrated remarkable capabilities in general video understanding, yet they often struggle with the fine-grained comprehension crucial for real-world applications requiring nuanced...

<https://arxiv.org/abs/2605.19846>

30 CONF

A Framework for Evaluating Zero-Shot Image Generation in Concept-based Explainability

ArXiv CS.AI 50%

arXiv:2605.19855v1 Announce Type: cross Abstract: Concept-based Explainable Artificial Intelligence (XAI) interprets deep learning models using human-understandable visual features (e.g., textures or object parts) by linking internal representations to class predictions, thereby...

<https://arxiv.org/abs/2605.19855>

31 CONF

Passive Construction Site Safety Monitoring via Persona-Scaffolded Adversarial Chain-of-Thought VLM Verification

ArXiv CS.AI 50%

arXiv:2605.19869v1 Announce Type: cross Abstract: Construction remains the deadliest industry sector in the United States, with 1,055 fatal worker injuries recorded in 2023, and the majority preventable. Existing monitoring approaches are expensive, require real-time human...

<https://arxiv.org/abs/2605.19869>

32 CONF

Deep Tech to Space: Space Data Centers and AI Revolution at the Edge

ArXiv CS.AI 50%

arXiv:2605.19892v1 Announce Type: cross Abstract: Dramatic cost reductions driven by private sector innovations have led to a rapid increase in the number of satellites in orbit and a corresponding surge in space-generated data. As this trend continues, transmitting large...

<https://arxiv.org/abs/2605.19892>

33 CONF

Fast and Featureless Node Representation Learning with Partial Pairwise Supervision

ArXiv CS.AI 50%

arXiv:2605.19916v1 Announce Type: cross Abstract: We introduce Contrastive FUSE, a fast and unified framework for scalable node representation learning in graphs with partially available pairwise node labels and no available node features. Unlike existing methods, we directly...

<https://arxiv.org/abs/2605.19916>

34 CONF

High-quality generation of dynamic game content via small language models: A proof of concept

ArXiv CS.AI 50%

arXiv:2601.23206v2 Announce Type: replace Abstract: Large language models (LLMs) offer promise for dynamic game content generation, but they face critical barriers, including narrative incoherence and high operational costs. Due to their large size, they are often accessed in...

<https://arxiv.org/abs/2601.23206>

35 CONF

Block-Sphere Vector Quantization

ArXiv CS.AI 50%

arXiv:2605.19972v1 Announce Type: cross Abstract: Vector quantization is a fundamental primitive for scalable machine learning systems, enabling memory-efficient storage, fast retrieval, and compressed inference. Recent rotation-based quantizers such as EDEN, RabbitQ, and...

<https://arxiv.org/abs/2605.19972>

36 CONF

LLM Benchmark Datasets Should Be Contamination-Resistant

ArXiv CS.AI 50%

arXiv:2605.19999v1 Announce Type: cross Abstract: Benchmark datasets are critical for reproducible, reliable, and discriminative evaluation of LLMs. However, recent studies reveal that many benchmark datasets are included in pretraining corpora, i.e., contaminated\$,...

<https://arxiv.org/abs/2605.19999>

37 CONF

Training Neural Networks with Optimal Double-Bayesian Learning

ArXiv CS.AI 50%

arXiv:2605.20009v1 Announce Type: cross Abstract: Backpropagation with gradient descent is a common optimization strategy employed by most neural network architectures in machine learning. However, finding optimal hyperparameters to guide training has proven challenging. While...

<https://arxiv.org/abs/2605.20009>

38 CONF

When Critics Disagree: Adaptive Reward Poisoning Attacks in RIS-Aided Wireless Control System

ArXiv CS.AI 50%

arXiv:2605.20037v1 Announce Type: cross Abstract: Reward-poisoning attacks present a significant risk to learning-based wireless control systems. Given this, we propose a Disagreement-Guided Reward Poisoning (DGRP) adaptive attack on a Soft Actor-Critic (SAC) agent. In a...

<https://arxiv.org/abs/2605.20037>

39 CONF

Does Code Cleanliness Affect Coding Agents? A Controlled Minimal-Pair Study

ArXiv CS.AI 50%

arXiv:2605.20049v1 Announce Type: cross Abstract: As autonomous coding agents see rapid adoption, their evaluation has primarily focused on task completion rates holding the target codebase fixed. This leaves a critical question unanswered: does the structural and stylistic...

<https://arxiv.org/abs/2605.20049>

40 CONF

Towards LLM-Assisted Architecture Recovery for Real-World ROS~2 Systems: An Agent-Based Multi-Level Approach to Hierarchical Structural Architecture Reconstruction

ArXiv CS.AI 50%

arXiv:2605.20055v1 Announce Type: cross Abstract: Explicit software architecture models are essential artifacts for communicating, analyzing, and evolving complex software-intensive systems. In ROS~2-based robotic systems, however, structural (de-)composition and integration...

<https://arxiv.org/abs/2605.20055>

41 CONF

Beyond Prediction Accuracy: Target-Space Recovery Profiles for Evaluating Model-Brain Alignment

ArXiv CS.AI 50%

arXiv:2605.20127v1 Announce Type: cross Abstract: Artificial vision models are often evaluated against the human visual cortex by measuring how accurately their internal representations predict brain responses. However, prediction accuracy alone does not indicate which...

<https://arxiv.org/abs/2605.20127>

42 CONF

Artificial Intelligence, conceptual metaphors and conceptual engineering: Are AI-based framings of human behaviour and cognition successful?

ArXiv CS.AI 50%

arXiv:2504.07756v2 Announce Type: replace Abstract: Understanding human behaviour, neuroscience and psychology using concepts from the domain of AI is increasing in popularity. Given the massive integration of AI technologies into our daily lives, AI-related concepts are being...

<https://arxiv.org/abs/2504.07756>

43 CONF

Nonlinearity as Rank: Generative Low-Rank Adapter with Radial Basis Functions

ArXiv CS.AI 50%

arXiv:2602.05709v2 Announce Type: replace Abstract: Low-rank adaptation (LoRA) approximates the update of a pretrained weight matrix using the product of two low-rank matrices. However, standard LoRA follows an explicit-rank paradigm, where increasing model capacity requires...

<https://arxiv.org/abs/2602.05709>

44 CONF

VERA-MH: Validation of Ethical and Responsible AI in Mental Health

ArXiv CS.AI 50%

arXiv:2605.13318v2 Announce Type: replace Abstract: Chatbot usage has increased, including in fields for which they were never developed for--notably mental health support. To that end, we introduce Validations of Ethical and Responsible AI in Mental Health (VERA-MH), a novel...

<https://arxiv.org/abs/2605.13318>

45 CONF

CBT-Audio: Evaluating Audio Language Models for Patient-Side Distress Intensity Estimation in CBT Session Recordings

ArXiv CS.AI 50%

arXiv:2605.17370v2 Announce Type: replace Abstract: Cognitive behavioural therapy is widely used to help patients understand and manage psychological distress. It is often delivered through spoken conversation, where therapists attend not only to what patients say, but also to...

<https://arxiv.org/abs/2605.17370>

46 CONF

Deep Neural Network for Musical Instrument Recognition using MFCCs

ArXiv CS.AI 50%

arXiv:2105.00933v3 Announce Type: replace-cross Abstract: The task of efficient automatic music classification is of vital importance and forms the basis for various advanced applications of AI in the musical domain. Musical instrument recognition is the task of instrument...

<https://arxiv.org/abs/2105.00933>

47 CONF

Monitoring AI-Modified Content at Scale: A Case Study on the Impact of ChatGPT on AI Conference Peer Reviews

ArXiv CS.AI 50%

arXiv:2403.07183v4 Announce Type: replace-cross Abstract: We present an approach for estimating the fraction of text in a large corpus which is likely to be substantially modified or produced by a large language model (LLM). Our maximum likelihood model leverages expert-written...

<https://arxiv.org/abs/2403.07183>

48 CONF

Slot-MLLM: Object-Centric Visual Tokenization for Multimodal LLM

ArXiv CS.AI 50%

arXiv:2505.17726v3 Announce Type: replace-cross Abstract: Recently, multimodal large language models (MLLMs) have emerged as a key approach in achieving artificial general intelligence. In particular, vision-language MLLMs have been developed to generate not only text but also...

<https://arxiv.org/abs/2505.17726>

49 CONF

Spatial-MLLM: Boosting MLLM Capabilities in Visual-based Spatial Intelligence

ArXiv CS.AI 50%

arXiv:2505.23747v2 Announce Type: replace-cross Abstract: Recent advancements in Multimodal Large Language Models (MLLMs) have significantly enhanced performance on 2D visual tasks. However, improving their spatial intelligence remains a challenge. Existing 3D MLLMs always rely...

<https://arxiv.org/abs/2505.23747>

50 CONF

Recursive Entropic Risk Optimization in Discounted MDPs: Sample Complexity Bounds with a Generative Model

ArXiv CS.AI 50%

arXiv:2506.00286v3 Announce Type: replace-cross Abstract: We study risk-sensitive reinforcement learning in finite discounted MDPs with recursive entropic risk measures (ERM), where the risk parameter $\beta \neq 0$ controls the agent's risk attitude: $\beta > 0$ for risk-averse...

<https://arxiv.org/abs/2506.00286>

51 CONF

YAC: Bridging Natural Language and Interactive Visual Exploration with Generative AI for Biomedical Data Discovery

ArXiv CS.AI 50%

arXiv:2509.19182v2 Announce Type: replace-cross Abstract: Incorporating natural language input has the potential to improve the capabilities of biomedical data discovery interfaces. However, user interface elements and visualizations are still powerful tools for interacting with...

<https://arxiv.org/abs/2509.19182>

52 CONF

Jailbreaking on Text-to-Video Models via Scene Splitting Strategy

ArXiv CS.AI 50%

arXiv:2509.22292v2 Announce Type: replace-cross Abstract: Along with the rapid advancement of numerous Text-to-Video (T2V) models, growing concerns have emerged regarding their safety risks. While recent studies have explored vulnerabilities in models like LLMs, VLMs, and...

<https://arxiv.org/abs/2509.22292>

53 CONF

HiDe: Rethinking The Zoom-IN method in High Resolution MLLMs via Hierarchical Decoupling

ArXiv CS.AI 50%

arXiv:2510.00054v2 Announce Type: replace-cross Abstract: Multimodal Large Language Models (MLLMs) have made significant strides in visual understanding tasks. However, their performance on high-resolution images remains suboptimal. While existing approaches often attribute this...

<https://arxiv.org/abs/2510.00054>

54 CONF

Beyond Majority Voting: LLM Aggregation by Leveraging Higher-Order Information

ArXiv CS.AI 50%

arXiv:2510.01499v2 Announce Type: replace-cross Abstract: With the rapid progress of multi-agent large language model (LLM) reasoning, how to effectively aggregate answers from multiple LLMs has emerged as a fundamental challenge. Standard majority voting treats all answers...

<https://arxiv.org/abs/2510.01499>

55 CONF

Proximal Diffusion Neural Sampler

ArXiv CS.AI 50%

arXiv:2510.03824v2 Announce Type: replace-cross Abstract: The task of learning a diffusion-based neural sampler for drawing samples from an unnormalized target distribution can be viewed as a stochastic optimal control problem on path measures. However, the training of neural...

<https://arxiv.org/abs/2510.03824>

56 CONF

Dr.LLM: Dynamic Layer Routing in LLMs

ArXiv CS.AI 50%

arXiv:2510.12773v2 Announce Type: replace-cross Abstract: Large Language Models (LLMs) process every token through all layers of a transformer stack, causing wasted computation on simple queries and insufficient flexibility for harder ones that need deeper reasoning....

<https://arxiv.org/abs/2510.12773>

57 CONF

HarmonicAttack: An Adaptive Cross-Domain Audio Watermark Removal

ArXiv CS.AI 50%

arXiv:2511.21577v2 Announce Type: replace-cross Abstract: The availability of high-quality, AI-generated audio raises security challenges such as misinformation campaigns and voice-cloning fraud. A key defense against the misuse of AI-generated audio is by watermarking it, so...

<https://arxiv.org/abs/2511.21577>

58 CONF

Open-Set Domain Adaptation Under Background Distribution Shift: Challenges and A Provably Efficient Solution

ArXiv CS.AI 50%

arXiv:2512.01152v4 Announce Type: replace-cross Abstract: As we deploy machine learning systems in the real world, a core challenge is to maintain a model that is performant even as the data shifts. Such shifts can take many forms: new classes may emerge that were absent during...

<https://arxiv.org/abs/2512.01152>

59 CONF

MaxShapley: Towards Incentive-compatible Generative Search with Fair Context Attribution

ArXiv CS.AI 50%

arXiv:2512.05958v2 Announce Type: replace-cross Abstract: Generative search engines based on large language models (LLMs) are replacing traditional search, fundamentally changing how information providers are compensated. To sustain this ecosystem, we need fair mechanisms to...

<https://arxiv.org/abs/2512.05958>

60 CONF

Toward Training Superintelligent Software Agents through Self-Play SWE-RL

ArXiv CS.AI 50%

arXiv:2512.18552v2 Announce Type: replace-cross Abstract: While current software agents powered by large language models (LLMs) and agentic reinforcement learning (RL) can boost programmer productivity, their training data (e.g., GitHub issues and pull requests) and environments...

<https://arxiv.org/abs/2512.18552>

61 CONF

FreeOrbit4D: Training-Free Arbitrary Camera Redirection for Monocular Videos via Foreground-Complete 4D Reconstruction

ArXiv CS.AI 50%

arXiv:2601.18993v2 Announce Type: replace-cross Abstract: Camera redirection aims to replay a dynamic scene from a single monocular video under a user-specified camera trajectory. However, large-angle redirection is inherently ill-posed: a monocular video captures only a narrow...

<https://arxiv.org/abs/2601.18993>

62 CONF

SuperInfer: SLO-Aware Rotary Scheduling and Memory Management for LLM Inference on Superchips

ArXiv CS.AI 50%

arXiv:2601.20309v2 Announce Type: replace-cross Abstract: Large Language Model (LLM) serving faces a fundamental tension between stringent latency Service Level Objectives (SLOs) and limited GPU memory capacity. When high request rates exhaust the KV cache budget, existing LLM...

<https://arxiv.org/abs/2601.20309>

63 CONF

WIND: Weather Inverse Diffusion for Zero-Shot Atmospheric Modeling

ArXiv CS.AI 50%

arXiv:2602.03924v2 Announce Type: replace-cross Abstract: Deep learning has revolutionized weather forecasting, but many challenges remain, including climate modeling. Moreover, the current landscape remains fragmented: highly specialized models are typically trained...

<https://arxiv.org/abs/2602.03924>

64 CONF

Learning Rate Matters: Vanilla LoRA May Suffice for LLM Fine-tuning

ArXiv CS.AI 50%

arXiv:2602.04998v2 Announce Type: replace-cross Abstract: Low-Rank Adaptation (LoRA) is the prevailing approach for efficient large language model (LLM) fine-tuning. Building on this paradigm, recent studies have proposed alternative initialization strategies, architectural...

<https://arxiv.org/abs/2602.04998>

65 CONF

How does longer temporal context enhance multimodal narrative video processing in the brain?

ArXiv CS.AI 50%

arXiv:2602.07570v2 Announce Type: replace-cross Abstract: Understanding how humans and artificial intelligence systems process complex narrative videos is a fundamental challenge at the intersection of neuroscience and machine learning. This study investigates how the temporal...

<https://arxiv.org/abs/2602.07570>

66 CONF

Language Model Memory and Memory Models for Language

ArXiv CS.AI 50%

arXiv:2602.13466v2 Announce Type: replace-cross Abstract: The ability of machine learning models to store input information in hidden layer vector embeddings, analogous to the concept of 'memory', is widely employed but not well characterized. We find that language model...

<https://arxiv.org/abs/2602.13466>

67 CONF

A Geometric Analysis of Small-sized Language Model Hallucinations

ArXiv CS.AI 50%

arXiv:2602.14778v3 Announce Type: replace-cross Abstract: Hallucinations -- plausible but factually incorrect responses -- pose a major challenge to the reliability of Large Language Models (LLMs), especially in multi-step or agentic settings. Existing work largely frames...

<https://arxiv.org/abs/2602.14778>

68 CONF

How to Model AI Agents as Personas?: Applying the Persona Ecosystem Playground to 41,300 Posts on Moltbook for Behavioral Insights

ArXiv CS.AI 50%

arXiv:2603.03140v3 Announce Type: replace-cross Abstract: AI agents are increasingly active on social media platforms, generating content and interacting with one another at scale. Yet the behavioral diversity of these agents remains poorly understood, and methods for...

<https://arxiv.org/abs/2603.03140>

69 CONF

Target-Aligned Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2603.29501v2 Announce Type: replace-cross Abstract: Many value-based deep reinforcement learning algorithms rely on target networks - lagged copies of the online network - to stabilize training. While effective, this mechanism introduces a fundamental stability-recency...

<https://arxiv.org/abs/2603.29501>

70 CONF

Addressing the Reality Gap: A Three-Tension Framework for Agentic AI Adoption

ArXiv CS.AI 50%

arXiv:2604.27245v2 Announce Type: replace-cross Abstract: Generative AI has rapidly entered education through free consumer tools, outpacing the ability of schools and universities to respond. Now a new wave of more autonomous agentic AI systems--with the capacity to plan and...

<https://arxiv.org/abs/2604.27245>

71 CONF

GEASS: Gated Evidence-Adaptive Selective Caption Trust for Vision-Language Models

ArXiv CS.AI 50%

arXiv:2605.01733v2 Announce Type: replace-cross Abstract: Vision-Language Models (VLMs) excel at grounded reasoning but remain prone to object hallucination. Recent work treats self-generated captions as a uniformly positive resource, yet we find that naively embedding one can...

<https://arxiv.org/abs/2605.01733>

72 CONF

VECTOR-Drive: Tightly Coupled Vision-Language and Trajectory Expert Routing for End-to-End Autonomous Driving

ArXiv CS.AI 50%

arXiv:2605.08830v2 Announce Type: replace-cross Abstract: End-to-end autonomous driving requires models to understand traffic scenes, infer driving intent, and generate executable motion plans. Recent vision-language-action (VLA) models inherit semantic priors from large-scale...

<https://arxiv.org/abs/2605.08830>

73 CONF

StrLoRA: Towards Streaming Continual Visual Instruction Tuning for MLLMs

ArXiv CS.AI 50%

arXiv:2605.16353v2 Announce Type: replace-cross Abstract: Continual Visual Instruction Tuning (CVIT) enables Multimodal Large Language Models to incrementally acquire new abilities. However, existing CVIT methods operate under a restrictive task-incremental setting, where each...

<https://arxiv.org/abs/2605.16353>